

```
In [1]: import numpy as np
import cv2
import os
import random
import matplotlib.pyplot as plt
import pickle
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.models import Model
import tensorflow as tf
from sklearn.metrics import classification_report, confusion_matrix
```

```
/opt/conda/lib/python3.10/site-packages/scipy/__init__.py:146: UserWarning: A NumPy
y version >=1.16.5 and <1.23.0 is required for this version of SciPy (detected ver
sion 1.23.5
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}")
```

```
In [2]: import tensorflow as tf
from tensorflow.keras import layers as L
import keras
from keras.layers import *
```

Data Preprocessing

```
In [3]: DIRECTORY = r'/kaggle/input/covid-ieee-train-test-split/COVID_IEEE Train Test Split'
CATEGORIES = ['covid', 'normal', 'virus']
IMG_SIZE = 224
train_data = []
```

```
In [4]: for category in CATEGORIES:
    folder = os.path.join(DIRECTORY, category)
    #print(folder)
    label = CATEGORIES.index(category)
    for img in os.listdir(folder):
        img_path = os.path.join(folder, img)
        #print(img_path)
        img_arr = cv2.imread(img_path)
        img_arr = cv2.resize(img_arr, (IMG_SIZE, IMG_SIZE))
        #plt.imshow(img_arr)
        #break
        train_data.append([img_arr, label])
```

```
In [5]: len(train_data)
```

```
Out[5]: 1457
```

```
In [6]: random.shuffle(train_data)
```

```
In [7]: X_train = []
y_train = []

for features, labels in train_data:
    X_train.append(features)
    y_train.append(labels)
```

```
In [8]: DIRECTORY = r'/kaggle/input/covid-ieee-train-test-split/COVID_IEEE Train Test Split'
CATEGORIES = ['covid', 'normal', 'virus']
```

```
IMG_SIZE = 224
test_data = []
```

```
In [9]: for category in CATEGORIES:
        folder = os.path.join(DIRECTORY, category)
        #print(folder)
        label = CATEGORIES.index(category)
        for img in os.listdir(folder):
            img_path = os.path.join(folder, img)
            #print(img_path)
            img_arr = cv2.imread(img_path)
            img_arr = cv2.resize(img_arr, (IMG_SIZE, IMG_SIZE))
            #plt.imshow(img_arr)
            #break
            test_data.append([img_arr, label])
```

```
In [10]: len(test_data)
```

```
Out[10]: 366
```

```
In [11]: X_test = []
        y_test = []

        for features, labels in test_data:
            X_test.append(features)
            y_test.append(labels)
```

```
In [12]: X_train = np.array(X_train)
        y_train = np.array(y_train)
        X_test = np.array(X_test)
        y_test = np.array(y_test)
```

```
In [13]: X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

```
Out[13]: ((1457, 224, 224, 3), (366, 224, 224, 3), (1457,), (366,))
```

Model

```
In [14]: data_augmentation = keras.Sequential(
        [
            Normalization(),
            RandomFlip("horizontal"),
            RandomRotation(factor=0.02),
            RandomZoom(
                height_factor=0.2, width_factor=0.2
            ),
            RandomContrast(0.2),
            RandomBrightness(0.2)
        ],
        name="data_augmentation",
    )
```

```
In [15]: class SpatialAttentionModule(tf.keras.layers.Layer):
        def __init__(self, kernel_size=3):
            """
            paper: https://arxiv.org/abs/1807.06521
            code: https://gist.github.com/innat/99888fa8065ecbf3ae2b297e5c10db70
            """
            super(SpatialAttentionModule, self).__init__()
```

```

self.conv1 = tf.keras.layers.Conv2D(64, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv2 = tf.keras.layers.Conv2D(32, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv3 = tf.keras.layers.Conv2D(16, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.nn.relu)

self.conv4 = tf.keras.layers.Conv2D(1, kernel_size=kernel_size,
                                     use_bias=False,
                                     kernel_initializer='he_normal',
                                     strides=1, padding='same',
                                     activation=tf.math.sigmoid)

def call(self, inputs):
    avg_out = tf.reduce_mean(inputs, axis=3)
    max_out = tf.reduce_max(inputs, axis=3)
    x = tf.stack([avg_out, max_out], axis=3)
    x = self.conv1(x)
    x = self.conv2(x)
    x = self.conv3(x)
    return self.conv4(x)

class ChannelAttentionModule(tf.keras.layers.Layer):
    def __init__(self, ratio=8):
        """
        paper: https://arxiv.org/abs/1807.06521
        code: https://gist.github.com/innat/99888fa8065ecbf3ae2b297e5c10db70
        """
        super(ChannelAttentionModule, self).__init__()
        self.ratio = ratio
        self.gapavg = tf.keras.layers.GlobalAveragePooling2D()
        self.gmpmax = tf.keras.layers.GlobalMaxPooling2D()

    def build(self, input_shape):
        self.conv1 = tf.keras.layers.Conv2D(input_shape[-1]//self.ratio,
                                             kernel_size=1,
                                             strides=1, padding='same',
                                             use_bias=True, activation=tf.nn.relu)

        self.conv2 = tf.keras.layers.Conv2D(input_shape[-1],
                                             kernel_size=1,
                                             strides=1, padding='same',
                                             use_bias=True, activation=tf.nn.relu)
        super(ChannelAttentionModule, self).build(input_shape)

    def call(self, inputs):
        # compute gap and gmp pooling
        gapavg = self.gapavg(inputs)
        gmpmax = self.gmpmax(inputs)
        gapavg = tf.keras.layers.Reshape((1, 1, gapavg.shape[1]))(gapavg)
        gmpmax = tf.keras.layers.Reshape((1, 1, gmpmax.shape[1]))(gmpmax)
        # forward passing to the respected layers
        gapavg_out = self.conv2(self.conv1(gapavg))
        gmpmax_out = self.conv2(self.conv1(gmpmax))
        return tf.math.sigmoid(gapavg_out + gmpmax_out)

```

```
def get_output_shape_for(self, input_shape):
    return self.compute_output_shape(input_shape)

def compute_output_shape(self, input_shape):
    output_len = input_shape[3]
    return (input_shape[0], output_len)
```

```
In [16]: def cbam(x):
x_ca = ChannelAttentionModule()(x)
x = Multiply()([x, x_ca])
x_sa = SpatialAttentionModule()(x)
x = Multiply()([x, x_sa])
return x
```

```
In [17]: def casa_module(x):
x_ca = ChannelAttentionModule()(x)
x_ca_out = Multiply()([x_ca, x])
x_sa = SpatialAttentionModule()(x)
x_sa_out = Multiply()([x_sa, x])
# Concatenate
x_casa = Concatenate()([x_ca_out, x_sa_out])
return x_casa
```

```
In [18]: def inverted_residual_block(inputs, num_filters, strides=1, expansion_ratio=1):
# point-wise conv
x = L.Conv2D(filters=expansion_ratio*inputs.shape[-1],
             kernel_size=1,
             padding='same',
             use_bias=False)(inputs)
x = L.BatchNormalization()(x)
x = L.Activation('swish')(x)

# depth-wise conv
x = L.DepthwiseConv2D(kernel_size=3,
                     strides=strides,
                     padding='same',
                     use_bias=False)(x)
x = L.BatchNormalization()(x)
x = L.Activation('swish')(x)

# point-wise conv
x = L.Conv2D(filters=num_filters,
             kernel_size=1,
             padding='same',
             use_bias=False)(x)
x = L.BatchNormalization()(x)

# Residual Connection
if strides == 1 and (inputs.shape == x.shape):
    return L.Add()([inputs, x])
return x
```

```
In [150... inputs = Input((224,224,3))
#x = ZeroPadding2D((3,3))(inputs)

x = SeparableConv2D(64, (3,3), strides=(2,2))(inputs)
x = BatchNormalization(axis=3)(x)
x = Activation('relu')(x)

x = SeparableConv2D(64, (3,3), activation='relu')(x)
```

```
x = MaxPooling2D((2,2))(x)
x = inverted_residual_block(x, 64, strides=1, expansion_ratio=2)

x1 = MaxPooling2D((2,2))(x)
x2 = casa_module(x)
x2 = MaxPooling2D((2,2))(x)
x = Concatenate()([x1, x2])

x = SeparableConv2D(128, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
x = SeparableConv2D(256, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
x = SeparableConv2D(256, (3,3), activation='relu')(x)
x = MaxPooling2D((2,2))(x)
```

In [151... `cnn = keras.Model(inputs, x)`

In [152... `cnn.summary()`

Model: "model_5"

Layer (type)	Output Shape	Param #	Connected to
=====			
input_27 (InputLayer)	[(None, 224, 224, 3)]	0	[]
separable_conv2d_25 (Separable [0]'] Conv2D)	(None, 111, 111, 64)	283	['input_27[0]
batch_normalization_24 (BatchN d_25[0][0]'] ormalization)	(None, 111, 111, 64)	256	['separable_conv2
activation_15 (Activation) tion_24[0][0]']	(None, 111, 111, 64)	0	['batch_normaliza
separable_conv2d_26 (Separable [0][0]'] Conv2D)	(None, 109, 109, 64)	4736	['activation_15
max_pooling2d_30 (MaxPooling2D d_26[0][0]'])	(None, 54, 54, 64)	0	['separable_conv2
conv2d_34 (Conv2D) 0[0][0]']	(None, 54, 54, 128)	8192	['max_pooling2d_3
batch_normalization_25 (BatchN [0]'] ormalization)	(None, 54, 54, 128)	512	['conv2d_34[0]
activation_16 (Activation) tion_25[0][0]']	(None, 54, 54, 128)	0	['batch_normaliza
depthwise_conv2d_5 (DepthwiseC [0][0]'] onv2D)	(None, 54, 54, 128)	1152	['activation_16
batch_normalization_26 (BatchN d_5[0][0]'] ormalization)	(None, 54, 54, 128)	512	['depthwise_conv2
activation_17 (Activation) tion_26[0][0]']	(None, 54, 54, 128)	0	['batch_normaliza
conv2d_35 (Conv2D) [0][0]']	(None, 54, 54, 64)	8192	['activation_17
batch_normalization_27 (BatchN [0]'] ormalization)	(None, 54, 54, 64)	256	['conv2d_35[0]
add_5 (Add) 0[0][0]',	(None, 54, 54, 64)	0	['max_pooling2d_3 'batch_normaliza
max_pooling2d_31 (MaxPooling2D)	(None, 27, 27, 64)	0	['add_5[0][0]']

```

max_pooling2d_32 (MaxPooling2D (None, 27, 27, 64) 0      ['add_5[0][0]']
)

concatenate_12 (Concatenate) (None, 27, 27, 128) 0      ['max_pooling2d_3
1[0][0]'],
                                     'max_pooling2d_3
2[0][0]']

separable_conv2d_27 (Separable (None, 25, 25, 128) 17664  ['concatenate_12
[0][0]']
Conv2D)

max_pooling2d_33 (MaxPooling2D (None, 12, 12, 128) 0      ['separable_conv2
d_27[0][0]']
)

separable_conv2d_28 (Separable (None, 10, 10, 256) 34176  ['max_pooling2d_3
3[0][0]']
Conv2D)

max_pooling2d_34 (MaxPooling2D (None, 5, 5, 256) 0      ['separable_conv2
d_28[0][0]']
)

separable_conv2d_29 (Separable (None, 3, 3, 256) 68096  ['max_pooling2d_3
4[0][0]']
Conv2D)

max_pooling2d_35 (MaxPooling2D (None, 1, 1, 256) 0      ['separable_conv2
d_29[0][0]']
)

=====
=====
Total params: 144,027
Trainable params: 143,259
Non-trainable params: 768

```

In [153...

```

model = tf.keras.Sequential([
    InputLayer((224,224,3)),
    data_augmentation,
    cnn,
    #Flatten(),
    GlobalAveragePooling2D(),
    #BatchNormalization(),
    Dropout(0.6),#0.75
    Dense(512, activation='relu'),
    Dropout(0.5),#0.75
    Dense(256, activation='relu'),
    #Dropout(0.2),#0.5
    Dense(128, activation='relu'),
    #Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(3, activation='softmax')
])

```

In [154...

```
model.summary()
```

Model: "sequential_21"

Layer (type)	Output Shape	Param #
data_augmentation (Sequential)	(None, 224, 224, 3)	7
model_5 (Functional)	(None, 1, 1, 256)	144027
global_average_pooling2d_28 (GlobalAveragePooling2D)	(None, 256)	0
dropout_53 (Dropout)	(None, 256)	0
dense_106 (Dense)	(None, 512)	131584
dropout_54 (Dropout)	(None, 512)	0
dense_107 (Dense)	(None, 256)	131328
dense_108 (Dense)	(None, 128)	32896
dense_109 (Dense)	(None, 32)	4128
dense_110 (Dense)	(None, 3)	99
Total params: 444,069		
Trainable params: 443,294		
Non-trainable params: 775		

```
In [155... class_weights = {0:1.1347,
                  1:0.9095,
                  2:0.9811}
```

```
In [156... from keras.callbacks import ModelCheckpoint, ReduceLROnPlateau
```

```
In [157... # Compile the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
In [158... filepath = "/kaggle/working/CASA_IRB_{epoch:02d}_{val_accuracy:.2f}.h5"
checkpoint = ModelCheckpoint(filepath, monitor="val_accuracy",
                             verbose=1, save_best_only=True,
                             mode='max')
reduce_lr = ReduceLROnPlateau(monitor='val_accuracy', factor=1e-1, patience=5, verbose=1)
callbacks = [checkpoint, reduce_lr]
```

```
In [159... history = model.fit(X_train, y_train,
                      validation_data=[X_test, y_test],
                      class_weight=class_weights,
                      callbacks=callbacks,
                      epochs=200, batch_size=32)
```


Epoch 1/200
45/46 [=====>.] - ETA: 0s - loss: 1.0983 - accuracy: 0.3521
Epoch 1: val_accuracy improved from -inf to 0.29508, saving model to /kaggle/worki
ng/CASA_IRB_01_0.30.h5
46/46 [=====] - 12s 88ms/step - loss: 1.0985 - accuracy:
0.3528 - val_loss: 1.0987 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 2/200
45/46 [=====>.] - ETA: 0s - loss: 1.0973 - accuracy: 0.4521
Epoch 2: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 72ms/step - loss: 1.0969 - accuracy:
0.4523 - val_loss: 1.0993 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 3/200
45/46 [=====>.] - ETA: 0s - loss: 1.0596 - accuracy: 0.5764
Epoch 3: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 1.0580 - accuracy:
0.5765 - val_loss: 1.1068 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 4/200
45/46 [=====>.] - ETA: 0s - loss: 0.8760 - accuracy: 0.6431
Epoch 4: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 73ms/step - loss: 0.8748 - accuracy:
0.6445 - val_loss: 1.1791 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 5/200
45/46 [=====>.] - ETA: 0s - loss: 0.7301 - accuracy: 0.6868
Epoch 5: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 0.7292 - accuracy:
0.6877 - val_loss: 1.4954 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 6/200
45/46 [=====>.] - ETA: 0s - loss: 0.6463 - accuracy: 0.7111
Epoch 6: val_accuracy did not improve from 0.29508

Epoch 6: ReduceLROnPlateau reducing learning rate to 5e-05.
46/46 [=====] - 3s 73ms/step - loss: 0.6452 - accuracy:
0.7131 - val_loss: 1.8163 - val_accuracy: 0.2951 - lr: 1.0000e-04
Epoch 7/200
45/46 [=====>.] - ETA: 0s - loss: 0.6109 - accuracy: 0.7271
Epoch 7: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 0.6096 - accuracy:
0.7282 - val_loss: 1.9237 - val_accuracy: 0.2951 - lr: 5.0000e-05
Epoch 8/200
45/46 [=====>.] - ETA: 0s - loss: 0.5919 - accuracy: 0.7597
Epoch 8: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 76ms/step - loss: 0.5909 - accuracy:
0.7598 - val_loss: 1.9923 - val_accuracy: 0.2951 - lr: 5.0000e-05
Epoch 9/200
45/46 [=====>.] - ETA: 0s - loss: 0.5701 - accuracy: 0.7486
Epoch 9: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 0.5706 - accuracy:
0.7495 - val_loss: 2.1010 - val_accuracy: 0.2951 - lr: 5.0000e-05
Epoch 10/200
45/46 [=====>.] - ETA: 0s - loss: 0.5576 - accuracy: 0.7576
Epoch 10: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 71ms/step - loss: 0.5563 - accuracy:
0.7577 - val_loss: 2.1363 - val_accuracy: 0.2951 - lr: 5.0000e-05
Epoch 11/200
45/46 [=====>.] - ETA: 0s - loss: 0.5356 - accuracy: 0.7833
Epoch 11: val_accuracy did not improve from 0.29508
46/46 [=====] - 3s 72ms/step - loss: 0.5339 - accuracy:
0.7838 - val_loss: 1.9497 - val_accuracy: 0.2951 - lr: 5.0000e-05
Epoch 12/200
45/46 [=====>.] - ETA: 0s - loss: 0.5097 - accuracy: 0.7875
Epoch 12: val_accuracy improved from 0.29508 to 0.32240, saving model to /kaggle/w
orking/CASA_IRB_12_0.32.h5
46/46 [=====] - 3s 76ms/step - loss: 0.5106 - accuracy:
0.7859 - val_loss: 1.5759 - val_accuracy: 0.3224 - lr: 5.0000e-05

Epoch 13/200
45/46 [=====>.] - ETA: 0s - loss: 0.5038 - accuracy: 0.8056
Epoch 13: val_accuracy improved from 0.32240 to 0.48907, saving model to /kaggle/working/CASA_IRB_13_0.49.h5
46/46 [=====] - 3s 74ms/step - loss: 0.5016 - accuracy: 0.8065 - val_loss: 0.9083 - val_accuracy: 0.4891 - lr: 5.0000e-05
Epoch 14/200
45/46 [=====>.] - ETA: 0s - loss: 0.4733 - accuracy: 0.8125
Epoch 14: val_accuracy improved from 0.48907 to 0.66940, saving model to /kaggle/working/CASA_IRB_14_0.67.h5
46/46 [=====] - 4s 76ms/step - loss: 0.4720 - accuracy: 0.8133 - val_loss: 0.6386 - val_accuracy: 0.6694 - lr: 5.0000e-05
Epoch 15/200
45/46 [=====>.] - ETA: 0s - loss: 0.4693 - accuracy: 0.8208
Epoch 15: val_accuracy improved from 0.66940 to 0.76776, saving model to /kaggle/working/CASA_IRB_15_0.77.h5
46/46 [=====] - 3s 76ms/step - loss: 0.4667 - accuracy: 0.8222 - val_loss: 0.4729 - val_accuracy: 0.7678 - lr: 5.0000e-05
Epoch 16/200
45/46 [=====>.] - ETA: 0s - loss: 0.4394 - accuracy: 0.8319
Epoch 16: val_accuracy improved from 0.76776 to 0.83333, saving model to /kaggle/working/CASA_IRB_16_0.83.h5
46/46 [=====] - 3s 74ms/step - loss: 0.4363 - accuracy: 0.8332 - val_loss: 0.3795 - val_accuracy: 0.8333 - lr: 5.0000e-05
Epoch 17/200
45/46 [=====>.] - ETA: 0s - loss: 0.4364 - accuracy: 0.8438
Epoch 17: val_accuracy improved from 0.83333 to 0.90984, saving model to /kaggle/working/CASA_IRB_17_0.91.h5
46/46 [=====] - 4s 78ms/step - loss: 0.4348 - accuracy: 0.8449 - val_loss: 0.2733 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 18/200
45/46 [=====>.] - ETA: 0s - loss: 0.3780 - accuracy: 0.8722
Epoch 18: val_accuracy did not improve from 0.90984
46/46 [=====] - 3s 75ms/step - loss: 0.3756 - accuracy: 0.8730 - val_loss: 0.2726 - val_accuracy: 0.8989 - lr: 5.0000e-05
Epoch 19/200
45/46 [=====>.] - ETA: 0s - loss: 0.3846 - accuracy: 0.8687
Epoch 19: val_accuracy improved from 0.90984 to 0.92350, saving model to /kaggle/working/CASA_IRB_19_0.92.h5
46/46 [=====] - 3s 76ms/step - loss: 0.3840 - accuracy: 0.8682 - val_loss: 0.2277 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 20/200
45/46 [=====>.] - ETA: 0s - loss: 0.3725 - accuracy: 0.8611
Epoch 20: val_accuracy improved from 0.92350 to 0.92623, saving model to /kaggle/working/CASA_IRB_20_0.93.h5
46/46 [=====] - 4s 77ms/step - loss: 0.3698 - accuracy: 0.8627 - val_loss: 0.2193 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 21/200
45/46 [=====>.] - ETA: 0s - loss: 0.3513 - accuracy: 0.8750
Epoch 21: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 71ms/step - loss: 0.3536 - accuracy: 0.8744 - val_loss: 0.2651 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 22/200
45/46 [=====>.] - ETA: 0s - loss: 0.3471 - accuracy: 0.8764
Epoch 22: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 72ms/step - loss: 0.3514 - accuracy: 0.8751 - val_loss: 0.2784 - val_accuracy: 0.9016 - lr: 5.0000e-05
Epoch 23/200
45/46 [=====>.] - ETA: 0s - loss: 0.3386 - accuracy: 0.8750
Epoch 23: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 74ms/step - loss: 0.3413 - accuracy: 0.8744 - val_loss: 0.2632 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 24/200
45/46 [=====>.] - ETA: 0s - loss: 0.3164 - accuracy: 0.8875

Epoch 24: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 74ms/step - loss: 0.3184 - accuracy: 0.8874 - val_loss: 0.2264 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 25/200
45/46 [=====>.] - ETA: 0s - loss: 0.3230 - accuracy: 0.8875
Epoch 25: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 74ms/step - loss: 0.3245 - accuracy: 0.8881 - val_loss: 0.2340 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 26/200
45/46 [=====>.] - ETA: 0s - loss: 0.3263 - accuracy: 0.8924
Epoch 26: val_accuracy did not improve from 0.92623
46/46 [=====] - 3s 71ms/step - loss: 0.3256 - accuracy: 0.8922 - val_loss: 0.2451 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 27/200
45/46 [=====>.] - ETA: 0s - loss: 0.3256 - accuracy: 0.8833
Epoch 27: val_accuracy improved from 0.92623 to 0.93169, saving model to /kaggle/working/CASA_IRB_27_0.93.h5
46/46 [=====] - 4s 79ms/step - loss: 0.3260 - accuracy: 0.8826 - val_loss: 0.2234 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 28/200
45/46 [=====>.] - ETA: 0s - loss: 0.3170 - accuracy: 0.8861
Epoch 28: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 74ms/step - loss: 0.3157 - accuracy: 0.8874 - val_loss: 0.2743 - val_accuracy: 0.8989 - lr: 5.0000e-05
Epoch 29/200
45/46 [=====>.] - ETA: 0s - loss: 0.3163 - accuracy: 0.8861
Epoch 29: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 72ms/step - loss: 0.3169 - accuracy: 0.8854 - val_loss: 0.2207 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 30/200
45/46 [=====>.] - ETA: 0s - loss: 0.3048 - accuracy: 0.8924
Epoch 30: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 73ms/step - loss: 0.3036 - accuracy: 0.8929 - val_loss: 0.2313 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 31/200
45/46 [=====>.] - ETA: 0s - loss: 0.2865 - accuracy: 0.9000
Epoch 31: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 73ms/step - loss: 0.2850 - accuracy: 0.8998 - val_loss: 0.2179 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 32/200
45/46 [=====>.] - ETA: 0s - loss: 0.2815 - accuracy: 0.8979
Epoch 32: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 74ms/step - loss: 0.2838 - accuracy: 0.8970 - val_loss: 0.2172 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 33/200
45/46 [=====>.] - ETA: 0s - loss: 0.2857 - accuracy: 0.8931
Epoch 33: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 73ms/step - loss: 0.2854 - accuracy: 0.8936 - val_loss: 0.2383 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 34/200
45/46 [=====>.] - ETA: 0s - loss: 0.2915 - accuracy: 0.8979
Epoch 34: val_accuracy did not improve from 0.93169
46/46 [=====] - 3s 74ms/step - loss: 0.2943 - accuracy: 0.8957 - val_loss: 0.2290 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 35/200
45/46 [=====>.] - ETA: 0s - loss: 0.2880 - accuracy: 0.9000
Epoch 35: val_accuracy improved from 0.93169 to 0.93443, saving model to /kaggle/working/CASA_IRB_35_0.93.h5
46/46 [=====] - 3s 74ms/step - loss: 0.2899 - accuracy: 0.8998 - val_loss: 0.2155 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 36/200
45/46 [=====>.] - ETA: 0s - loss: 0.2788 - accuracy: 0.8986
Epoch 36: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2791 - accuracy:

0.8984 - val_loss: 0.2578 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 37/200
45/46 [=====>.] - ETA: 0s - loss: 0.2656 - accuracy: 0.9035
Epoch 37: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2668 - accuracy: 0.9039 - val_loss: 0.2522 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 38/200
45/46 [=====>.] - ETA: 0s - loss: 0.2817 - accuracy: 0.9021
Epoch 38: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 72ms/step - loss: 0.2818 - accuracy: 0.9025 - val_loss: 0.2359 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 39/200
45/46 [=====>.] - ETA: 0s - loss: 0.2726 - accuracy: 0.8958
Epoch 39: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 71ms/step - loss: 0.2756 - accuracy: 0.8950 - val_loss: 0.2423 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 40/200
45/46 [=====>.] - ETA: 0s - loss: 0.2688 - accuracy: 0.9111
Epoch 40: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2690 - accuracy: 0.9115 - val_loss: 0.2355 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 41/200
45/46 [=====>.] - ETA: 0s - loss: 0.2897 - accuracy: 0.9014
Epoch 41: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2883 - accuracy: 0.9019 - val_loss: 0.2612 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 42/200
45/46 [=====>.] - ETA: 0s - loss: 0.2660 - accuracy: 0.9104
Epoch 42: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2642 - accuracy: 0.9108 - val_loss: 0.2242 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 43/200
45/46 [=====>.] - ETA: 0s - loss: 0.2670 - accuracy: 0.9090
Epoch 43: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 71ms/step - loss: 0.2663 - accuracy: 0.9087 - val_loss: 0.3435 - val_accuracy: 0.8607 - lr: 5.0000e-05
Epoch 44/200
45/46 [=====>.] - ETA: 0s - loss: 0.2750 - accuracy: 0.9042
Epoch 44: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2738 - accuracy: 0.9039 - val_loss: 0.2257 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 45/200
45/46 [=====>.] - ETA: 0s - loss: 0.2764 - accuracy: 0.9000
Epoch 45: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 72ms/step - loss: 0.2775 - accuracy: 0.8998 - val_loss: 0.2445 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 46/200
45/46 [=====>.] - ETA: 0s - loss: 0.2369 - accuracy: 0.9208
Epoch 46: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2352 - accuracy: 0.9218 - val_loss: 0.2252 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 47/200
45/46 [=====>.] - ETA: 0s - loss: 0.2551 - accuracy: 0.9076
Epoch 47: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 74ms/step - loss: 0.2592 - accuracy: 0.9073 - val_loss: 0.2788 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 48/200
45/46 [=====>.] - ETA: 0s - loss: 0.2606 - accuracy: 0.9062
Epoch 48: val_accuracy did not improve from 0.93443
46/46 [=====] - 3s 71ms/step - loss: 0.2601 - accuracy: 0.9073 - val_loss: 0.3103 - val_accuracy: 0.8743 - lr: 5.0000e-05
Epoch 49/200
45/46 [=====>.] - ETA: 0s - loss: 0.2484 - accuracy: 0.9146
Epoch 49: val_accuracy improved from 0.93443 to 0.94262, saving model to /kaggle/w

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orking/CASA_IRB_49_0.94.h5
46/46 [=====] - 3s 74ms/step - loss: 0.2476 - accuracy:
0.9142 - val_loss: 0.2197 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 50/200
45/46 [=====>.] - ETA: 0s - loss: 0.2658 - accuracy: 0.9083
Epoch 50: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2702 - accuracy:
0.9067 - val_loss: 0.2562 - val_accuracy: 0.9016 - lr: 5.0000e-05
Epoch 51/200
45/46 [=====>.] - ETA: 0s - loss: 0.2579 - accuracy: 0.9125
Epoch 51: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2593 - accuracy:
0.9121 - val_loss: 0.2616 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 52/200
45/46 [=====>.] - ETA: 0s - loss: 0.2566 - accuracy: 0.9097
Epoch 52: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2586 - accuracy:
0.9087 - val_loss: 0.3082 - val_accuracy: 0.8798 - lr: 5.0000e-05
Epoch 53/200
45/46 [=====>.] - ETA: 0s - loss: 0.2292 - accuracy: 0.9215
Epoch 53: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.2276 - accuracy:
0.9224 - val_loss: 0.2082 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 54/200
45/46 [=====>.] - ETA: 0s - loss: 0.2619 - accuracy: 0.9104
Epoch 54: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2591 - accuracy:
0.9115 - val_loss: 0.2668 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 55/200
45/46 [=====>.] - ETA: 0s - loss: 0.2420 - accuracy: 0.9174
Epoch 55: val_accuracy did not improve from 0.94262
46/46 [=====] - 4s 77ms/step - loss: 0.2401 - accuracy:
0.9183 - val_loss: 0.2868 - val_accuracy: 0.8907 - lr: 5.0000e-05
Epoch 56/200
45/46 [=====>.] - ETA: 0s - loss: 0.2439 - accuracy: 0.9118
Epoch 56: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2468 - accuracy:
0.9121 - val_loss: 0.2414 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 57/200
45/46 [=====>.] - ETA: 0s - loss: 0.2565 - accuracy: 0.9132
Epoch 57: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2607 - accuracy:
0.9121 - val_loss: 0.2472 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 58/200
45/46 [=====>.] - ETA: 0s - loss: 0.2471 - accuracy: 0.9118
Epoch 58: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2490 - accuracy:
0.9108 - val_loss: 0.3063 - val_accuracy: 0.8907 - lr: 5.0000e-05
Epoch 59/200
45/46 [=====>.] - ETA: 0s - loss: 0.2502 - accuracy: 0.9201
Epoch 59: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.2495 - accuracy:
0.9197 - val_loss: 0.2773 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 60/200
45/46 [=====>.] - ETA: 0s - loss: 0.2412 - accuracy: 0.9174
Epoch 60: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2411 - accuracy:
0.9170 - val_loss: 0.2425 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 61/200
45/46 [=====>.] - ETA: 0s - loss: 0.2400 - accuracy: 0.9125
Epoch 61: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.2396 - accuracy:
0.9128 - val_loss: 0.2615 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 62/200
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45/46 [=====>.] - ETA: 0s - loss: 0.2389 - accuracy: 0.9236
Epoch 62: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2382 - accuracy: 0.9231 - val_loss: 0.2464 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 63/200
45/46 [=====>.] - ETA: 0s - loss: 0.2232 - accuracy: 0.9215
Epoch 63: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2227 - accuracy: 0.9218 - val_loss: 0.2946 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 64/200
45/46 [=====>.] - ETA: 0s - loss: 0.2371 - accuracy: 0.9132
Epoch 64: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2353 - accuracy: 0.9142 - val_loss: 0.2740 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 65/200
45/46 [=====>.] - ETA: 0s - loss: 0.2310 - accuracy: 0.9194
Epoch 65: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2301 - accuracy: 0.9197 - val_loss: 0.2443 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 66/200
45/46 [=====>.] - ETA: 0s - loss: 0.2362 - accuracy: 0.9132
Epoch 66: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2340 - accuracy: 0.9142 - val_loss: 0.2894 - val_accuracy: 0.8934 - lr: 5.0000e-05
Epoch 67/200
45/46 [=====>.] - ETA: 0s - loss: 0.2416 - accuracy: 0.9181
Epoch 67: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2395 - accuracy: 0.9190 - val_loss: 0.2773 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 68/200
45/46 [=====>.] - ETA: 0s - loss: 0.2431 - accuracy: 0.9222
Epoch 68: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2437 - accuracy: 0.9211 - val_loss: 0.2565 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 69/200
45/46 [=====>.] - ETA: 0s - loss: 0.2312 - accuracy: 0.9187
Epoch 69: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2314 - accuracy: 0.9183 - val_loss: 0.3373 - val_accuracy: 0.8579 - lr: 5.0000e-05
Epoch 70/200
45/46 [=====>.] - ETA: 0s - loss: 0.2381 - accuracy: 0.9236
Epoch 70: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2403 - accuracy: 0.9231 - val_loss: 0.2677 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 71/200
45/46 [=====>.] - ETA: 0s - loss: 0.2257 - accuracy: 0.9215
Epoch 71: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2245 - accuracy: 0.9218 - val_loss: 0.2585 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 72/200
45/46 [=====>.] - ETA: 0s - loss: 0.2257 - accuracy: 0.9153
Epoch 72: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2285 - accuracy: 0.9149 - val_loss: 0.2434 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 73/200
45/46 [=====>.] - ETA: 0s - loss: 0.2324 - accuracy: 0.9194
Epoch 73: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2351 - accuracy: 0.9190 - val_loss: 0.2642 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 74/200
45/46 [=====>.] - ETA: 0s - loss: 0.2226 - accuracy: 0.9222
Epoch 74: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2208 - accuracy: 0.9231 - val_loss: 0.2452 - val_accuracy: 0.9290 - lr: 5.0000e-05

Epoch 75/200
45/46 [=====>.] - ETA: 0s - loss: 0.2214 - accuracy: 0.9236
Epoch 75: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.2192 - accuracy: 0.9245 - val_loss: 0.2682 - val_accuracy: 0.9016 - lr: 5.0000e-05
Epoch 76/200
45/46 [=====>.] - ETA: 0s - loss: 0.2271 - accuracy: 0.9181
Epoch 76: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2257 - accuracy: 0.9183 - val_loss: 0.2469 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 77/200
45/46 [=====>.] - ETA: 0s - loss: 0.2217 - accuracy: 0.9229
Epoch 77: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2198 - accuracy: 0.9238 - val_loss: 0.2381 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 78/200
45/46 [=====>.] - ETA: 0s - loss: 0.2285 - accuracy: 0.9264
Epoch 78: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2279 - accuracy: 0.9266 - val_loss: 0.2546 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 79/200
45/46 [=====>.] - ETA: 0s - loss: 0.2339 - accuracy: 0.9236
Epoch 79: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2327 - accuracy: 0.9238 - val_loss: 0.3157 - val_accuracy: 0.8689 - lr: 5.0000e-05
Epoch 80/200
45/46 [=====>.] - ETA: 0s - loss: 0.2247 - accuracy: 0.9222
Epoch 80: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2227 - accuracy: 0.9231 - val_loss: 0.2496 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 81/200
45/46 [=====>.] - ETA: 0s - loss: 0.2240 - accuracy: 0.9264
Epoch 81: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2245 - accuracy: 0.9259 - val_loss: 0.2776 - val_accuracy: 0.8934 - lr: 5.0000e-05
Epoch 82/200
45/46 [=====>.] - ETA: 0s - loss: 0.2254 - accuracy: 0.9201
Epoch 82: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2232 - accuracy: 0.9211 - val_loss: 0.2997 - val_accuracy: 0.8798 - lr: 5.0000e-05
Epoch 83/200
45/46 [=====>.] - ETA: 0s - loss: 0.2192 - accuracy: 0.9250
Epoch 83: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2189 - accuracy: 0.9252 - val_loss: 0.2816 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 84/200
45/46 [=====>.] - ETA: 0s - loss: 0.2183 - accuracy: 0.9257
Epoch 84: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 76ms/step - loss: 0.2195 - accuracy: 0.9245 - val_loss: 0.2631 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 85/200
45/46 [=====>.] - ETA: 0s - loss: 0.2235 - accuracy: 0.9160
Epoch 85: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2248 - accuracy: 0.9163 - val_loss: 0.3022 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 86/200
45/46 [=====>.] - ETA: 0s - loss: 0.1987 - accuracy: 0.9340
Epoch 86: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2027 - accuracy: 0.9327 - val_loss: 0.3025 - val_accuracy: 0.8798 - lr: 5.0000e-05
Epoch 87/200
45/46 [=====>.] - ETA: 0s - loss: 0.2216 - accuracy: 0.9222
Epoch 87: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2222 - accuracy:

0.9211 - val_loss: 0.2294 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 88/200
45/46 [=====>.] - ETA: 0s - loss: 0.2140 - accuracy: 0.9306
Epoch 88: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2131 - accuracy: 0.9314 - val_loss: 0.2480 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 89/200
45/46 [=====>.] - ETA: 0s - loss: 0.2059 - accuracy: 0.9306
Epoch 89: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2066 - accuracy: 0.9307 - val_loss: 0.2987 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 90/200
45/46 [=====>.] - ETA: 0s - loss: 0.2296 - accuracy: 0.9271
Epoch 90: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2283 - accuracy: 0.9272 - val_loss: 0.3525 - val_accuracy: 0.8579 - lr: 5.0000e-05
Epoch 91/200
45/46 [=====>.] - ETA: 0s - loss: 0.2032 - accuracy: 0.9319
Epoch 91: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.2025 - accuracy: 0.9321 - val_loss: 0.2550 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 92/200
45/46 [=====>.] - ETA: 0s - loss: 0.2150 - accuracy: 0.9292
Epoch 92: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2194 - accuracy: 0.9272 - val_loss: 0.3331 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 93/200
45/46 [=====>.] - ETA: 0s - loss: 0.2031 - accuracy: 0.9285
Epoch 93: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 76ms/step - loss: 0.2027 - accuracy: 0.9279 - val_loss: 0.2795 - val_accuracy: 0.8934 - lr: 5.0000e-05
Epoch 94/200
45/46 [=====>.] - ETA: 0s - loss: 0.2161 - accuracy: 0.9194
Epoch 94: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2167 - accuracy: 0.9190 - val_loss: 0.2297 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 95/200
45/46 [=====>.] - ETA: 0s - loss: 0.2055 - accuracy: 0.9243
Epoch 95: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2062 - accuracy: 0.9238 - val_loss: 0.2342 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 96/200
45/46 [=====>.] - ETA: 0s - loss: 0.2110 - accuracy: 0.9319
Epoch 96: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2095 - accuracy: 0.9327 - val_loss: 0.2266 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 97/200
45/46 [=====>.] - ETA: 0s - loss: 0.2093 - accuracy: 0.9278
Epoch 97: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2131 - accuracy: 0.9266 - val_loss: 0.2935 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 98/200
45/46 [=====>.] - ETA: 0s - loss: 0.1835 - accuracy: 0.9389
Epoch 98: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1819 - accuracy: 0.9396 - val_loss: 0.2271 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 99/200
45/46 [=====>.] - ETA: 0s - loss: 0.1979 - accuracy: 0.9347
Epoch 99: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1976 - accuracy: 0.9348 - val_loss: 0.2377 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 100/200
45/46 [=====>.] - ETA: 0s - loss: 0.2111 - accuracy: 0.9257
Epoch 100: val_accuracy did not improve from 0.94262

46/46 [=====] - 3s 72ms/step - loss: 0.2100 - accuracy: 0.9266 - val_loss: 0.2357 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 101/200
45/46 [=====>.] - ETA: 0s - loss: 0.2081 - accuracy: 0.9361
Epoch 101: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2103 - accuracy: 0.9355 - val_loss: 0.4047 - val_accuracy: 0.8443 - lr: 5.0000e-05
Epoch 102/200
45/46 [=====>.] - ETA: 0s - loss: 0.2049 - accuracy: 0.9306
Epoch 102: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2033 - accuracy: 0.9314 - val_loss: 0.2300 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 103/200
45/46 [=====>.] - ETA: 0s - loss: 0.2004 - accuracy: 0.9306
Epoch 103: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1986 - accuracy: 0.9314 - val_loss: 0.2930 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 104/200
45/46 [=====>.] - ETA: 0s - loss: 0.2116 - accuracy: 0.9271
Epoch 104: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2117 - accuracy: 0.9272 - val_loss: 0.2567 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 105/200
45/46 [=====>.] - ETA: 0s - loss: 0.2074 - accuracy: 0.9326
Epoch 105: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2057 - accuracy: 0.9334 - val_loss: 0.2551 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 106/200
45/46 [=====>.] - ETA: 0s - loss: 0.2051 - accuracy: 0.9319
Epoch 106: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2042 - accuracy: 0.9327 - val_loss: 0.3261 - val_accuracy: 0.8661 - lr: 5.0000e-05
Epoch 107/200
45/46 [=====>.] - ETA: 0s - loss: 0.2014 - accuracy: 0.9299
Epoch 107: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2010 - accuracy: 0.9293 - val_loss: 0.2442 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 108/200
45/46 [=====>.] - ETA: 0s - loss: 0.1934 - accuracy: 0.9319
Epoch 108: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1948 - accuracy: 0.9321 - val_loss: 0.2577 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 109/200
45/46 [=====>.] - ETA: 0s - loss: 0.2074 - accuracy: 0.9312
Epoch 109: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.2057 - accuracy: 0.9321 - val_loss: 0.2899 - val_accuracy: 0.8907 - lr: 5.0000e-05
Epoch 110/200
45/46 [=====>.] - ETA: 0s - loss: 0.1924 - accuracy: 0.9312
Epoch 110: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1914 - accuracy: 0.9314 - val_loss: 0.4572 - val_accuracy: 0.8279 - lr: 5.0000e-05
Epoch 111/200
45/46 [=====>.] - ETA: 0s - loss: 0.1939 - accuracy: 0.9382
Epoch 111: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1922 - accuracy: 0.9389 - val_loss: 0.3511 - val_accuracy: 0.8689 - lr: 5.0000e-05
Epoch 112/200
45/46 [=====>.] - ETA: 0s - loss: 0.1989 - accuracy: 0.9306
Epoch 112: val_accuracy did not improve from 0.94262
46/46 [=====] - 4s 77ms/step - loss: 0.1983 - accuracy: 0.9307 - val_loss: 0.3385 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 113/200
45/46 [=====>.] - ETA: 0s - loss: 0.1919 - accuracy: 0.9319

Epoch 113: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1939 - accuracy: 0.9307 - val_loss: 0.2915 - val_accuracy: 0.8880 - lr: 5.0000e-05
Epoch 114/200
45/46 [=====>.] - ETA: 0s - loss: 0.1947 - accuracy: 0.9319
Epoch 114: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1929 - accuracy: 0.9327 - val_loss: 0.2547 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 115/200
45/46 [=====>.] - ETA: 0s - loss: 0.2062 - accuracy: 0.9292
Epoch 115: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.2052 - accuracy: 0.9293 - val_loss: 0.3684 - val_accuracy: 0.8579 - lr: 5.0000e-05
Epoch 116/200
45/46 [=====>.] - ETA: 0s - loss: 0.1874 - accuracy: 0.9354
Epoch 116: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1872 - accuracy: 0.9355 - val_loss: 0.2905 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 117/200
45/46 [=====>.] - ETA: 0s - loss: 0.2033 - accuracy: 0.9271
Epoch 117: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.2078 - accuracy: 0.9266 - val_loss: 0.2616 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 118/200
45/46 [=====>.] - ETA: 0s - loss: 0.1870 - accuracy: 0.9389
Epoch 118: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1878 - accuracy: 0.9389 - val_loss: 0.2852 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 119/200
45/46 [=====>.] - ETA: 0s - loss: 0.1800 - accuracy: 0.9444
Epoch 119: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1807 - accuracy: 0.9437 - val_loss: 0.3050 - val_accuracy: 0.8907 - lr: 5.0000e-05
Epoch 120/200
45/46 [=====>.] - ETA: 0s - loss: 0.1928 - accuracy: 0.9264
Epoch 120: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1917 - accuracy: 0.9272 - val_loss: 0.3619 - val_accuracy: 0.8579 - lr: 5.0000e-05
Epoch 121/200
45/46 [=====>.] - ETA: 0s - loss: 0.1834 - accuracy: 0.9382
Epoch 121: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1862 - accuracy: 0.9369 - val_loss: 0.4103 - val_accuracy: 0.8361 - lr: 5.0000e-05
Epoch 122/200
45/46 [=====>.] - ETA: 0s - loss: 0.1933 - accuracy: 0.9347
Epoch 122: val_accuracy did not improve from 0.94262
46/46 [=====] - 4s 77ms/step - loss: 0.1939 - accuracy: 0.9341 - val_loss: 0.3100 - val_accuracy: 0.8880 - lr: 5.0000e-05
Epoch 123/200
45/46 [=====>.] - ETA: 0s - loss: 0.1884 - accuracy: 0.9375
Epoch 123: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1909 - accuracy: 0.9369 - val_loss: 0.2659 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 124/200
45/46 [=====>.] - ETA: 0s - loss: 0.1757 - accuracy: 0.9410
Epoch 124: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1778 - accuracy: 0.9410 - val_loss: 0.2373 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 125/200
45/46 [=====>.] - ETA: 0s - loss: 0.1777 - accuracy: 0.9410
Epoch 125: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1761 - accuracy: 0.9417 - val_loss: 0.2575 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 126/200

45/46 [=====>.] - ETA: 0s - loss: 0.1840 - accuracy: 0.9347
Epoch 126: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1824 - accuracy: 0.9355 - val_loss: 0.2580 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 127/200
45/46 [=====>.] - ETA: 0s - loss: 0.1932 - accuracy: 0.9333
Epoch 127: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1941 - accuracy: 0.9327 - val_loss: 0.2472 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 128/200
45/46 [=====>.] - ETA: 0s - loss: 0.1753 - accuracy: 0.9431
Epoch 128: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1742 - accuracy: 0.9437 - val_loss: 0.2688 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 129/200
45/46 [=====>.] - ETA: 0s - loss: 0.1912 - accuracy: 0.9299
Epoch 129: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1918 - accuracy: 0.9300 - val_loss: 0.2380 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 130/200
45/46 [=====>.] - ETA: 0s - loss: 0.1891 - accuracy: 0.9326
Epoch 130: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1879 - accuracy: 0.9327 - val_loss: 0.2637 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 131/200
45/46 [=====>.] - ETA: 0s - loss: 0.1836 - accuracy: 0.9403
Epoch 131: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 76ms/step - loss: 0.1837 - accuracy: 0.9403 - val_loss: 0.3232 - val_accuracy: 0.8798 - lr: 5.0000e-05
Epoch 132/200
45/46 [=====>.] - ETA: 0s - loss: 0.1921 - accuracy: 0.9347
Epoch 132: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1939 - accuracy: 0.9334 - val_loss: 0.2826 - val_accuracy: 0.8880 - lr: 5.0000e-05
Epoch 133/200
45/46 [=====>.] - ETA: 0s - loss: 0.1759 - accuracy: 0.9368
Epoch 133: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1746 - accuracy: 0.9375 - val_loss: 0.2791 - val_accuracy: 0.8880 - lr: 5.0000e-05
Epoch 134/200
45/46 [=====>.] - ETA: 0s - loss: 0.1838 - accuracy: 0.9382
Epoch 134: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1824 - accuracy: 0.9389 - val_loss: 0.3381 - val_accuracy: 0.8661 - lr: 5.0000e-05
Epoch 135/200
45/46 [=====>.] - ETA: 0s - loss: 0.1805 - accuracy: 0.9396
Epoch 135: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1793 - accuracy: 0.9396 - val_loss: 0.2773 - val_accuracy: 0.8907 - lr: 5.0000e-05
Epoch 136/200
45/46 [=====>.] - ETA: 0s - loss: 0.1716 - accuracy: 0.9438
Epoch 136: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1720 - accuracy: 0.9430 - val_loss: 0.2768 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 137/200
45/46 [=====>.] - ETA: 0s - loss: 0.1849 - accuracy: 0.9396
Epoch 137: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1856 - accuracy: 0.9396 - val_loss: 0.2511 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 138/200
45/46 [=====>.] - ETA: 0s - loss: 0.1771 - accuracy: 0.9444
Epoch 138: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1772 - accuracy: 0.9444 - val_loss: 0.3079 - val_accuracy: 0.8770 - lr: 5.0000e-05

Epoch 139/200
45/46 [=====>.] - ETA: 0s - loss: 0.1663 - accuracy: 0.9458
Epoch 139: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1647 - accuracy: 0.9465 - val_loss: 0.2970 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 140/200
45/46 [=====>.] - ETA: 0s - loss: 0.1713 - accuracy: 0.9403
Epoch 140: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1702 - accuracy: 0.9410 - val_loss: 0.3112 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 141/200
45/46 [=====>.] - ETA: 0s - loss: 0.1817 - accuracy: 0.9361
Epoch 141: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1810 - accuracy: 0.9362 - val_loss: 0.3564 - val_accuracy: 0.8607 - lr: 5.0000e-05
Epoch 142/200
45/46 [=====>.] - ETA: 0s - loss: 0.1804 - accuracy: 0.9354
Epoch 142: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1817 - accuracy: 0.9355 - val_loss: 0.2536 - val_accuracy: 0.9262 - lr: 5.0000e-05
Epoch 143/200
45/46 [=====>.] - ETA: 0s - loss: 0.1666 - accuracy: 0.9410
Epoch 143: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1676 - accuracy: 0.9403 - val_loss: 0.3000 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 144/200
45/46 [=====>.] - ETA: 0s - loss: 0.1673 - accuracy: 0.9451
Epoch 144: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1716 - accuracy: 0.9451 - val_loss: 0.2611 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 145/200
45/46 [=====>.] - ETA: 0s - loss: 0.1915 - accuracy: 0.9319
Epoch 145: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1898 - accuracy: 0.9327 - val_loss: 0.3047 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 146/200
45/46 [=====>.] - ETA: 0s - loss: 0.1681 - accuracy: 0.9465
Epoch 146: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1686 - accuracy: 0.9465 - val_loss: 0.2725 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 147/200
45/46 [=====>.] - ETA: 0s - loss: 0.1699 - accuracy: 0.9444
Epoch 147: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1713 - accuracy: 0.9437 - val_loss: 0.3186 - val_accuracy: 0.8770 - lr: 5.0000e-05
Epoch 148/200
45/46 [=====>.] - ETA: 0s - loss: 0.1717 - accuracy: 0.9424
Epoch 148: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1712 - accuracy: 0.9423 - val_loss: 0.2678 - val_accuracy: 0.9126 - lr: 5.0000e-05
Epoch 149/200
45/46 [=====>.] - ETA: 0s - loss: 0.1517 - accuracy: 0.9514
Epoch 149: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1513 - accuracy: 0.9513 - val_loss: 0.2317 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 150/200
45/46 [=====>.] - ETA: 0s - loss: 0.1865 - accuracy: 0.9431
Epoch 150: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 76ms/step - loss: 0.1863 - accuracy: 0.9430 - val_loss: 0.2465 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 151/200
46/46 [=====] - ETA: 0s - loss: 0.1661 - accuracy: 0.9451
Epoch 151: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1661 - accuracy:

0.9451 - val_loss: 0.2619 - val_accuracy: 0.9098 - lr: 5.0000e-05
Epoch 152/200
45/46 [=====>.] - ETA: 0s - loss: 0.1697 - accuracy: 0.9431
Epoch 152: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1684 - accuracy: 0.9437 - val_loss: 0.2345 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 153/200
45/46 [=====>.] - ETA: 0s - loss: 0.1790 - accuracy: 0.9431
Epoch 153: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1803 - accuracy: 0.9417 - val_loss: 0.2935 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 154/200
46/46 [=====] - ETA: 0s - loss: 0.1653 - accuracy: 0.9396
Epoch 154: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1653 - accuracy: 0.9396 - val_loss: 0.2665 - val_accuracy: 0.8989 - lr: 5.0000e-05
Epoch 155/200
45/46 [=====>.] - ETA: 0s - loss: 0.1733 - accuracy: 0.9417
Epoch 155: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1718 - accuracy: 0.9423 - val_loss: 0.2306 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 156/200
45/46 [=====>.] - ETA: 0s - loss: 0.1694 - accuracy: 0.9451
Epoch 156: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1684 - accuracy: 0.9451 - val_loss: 0.2432 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 157/200
46/46 [=====] - ETA: 0s - loss: 0.1804 - accuracy: 0.9403
Epoch 157: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 75ms/step - loss: 0.1804 - accuracy: 0.9403 - val_loss: 0.2285 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 158/200
45/46 [=====>.] - ETA: 0s - loss: 0.1893 - accuracy: 0.9333
Epoch 158: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 73ms/step - loss: 0.1896 - accuracy: 0.9327 - val_loss: 0.2554 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 159/200
45/46 [=====>.] - ETA: 0s - loss: 0.1683 - accuracy: 0.9507
Epoch 159: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 72ms/step - loss: 0.1690 - accuracy: 0.9506 - val_loss: 0.2550 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 160/200
45/46 [=====>.] - ETA: 0s - loss: 0.1629 - accuracy: 0.9410
Epoch 160: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 75ms/step - loss: 0.1616 - accuracy: 0.9417 - val_loss: 0.3505 - val_accuracy: 0.8689 - lr: 5.0000e-05
Epoch 161/200
45/46 [=====>.] - ETA: 0s - loss: 0.1847 - accuracy: 0.9382
Epoch 161: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1835 - accuracy: 0.9389 - val_loss: 0.3209 - val_accuracy: 0.8716 - lr: 5.0000e-05
Epoch 162/200
45/46 [=====>.] - ETA: 0s - loss: 0.1621 - accuracy: 0.9472
Epoch 162: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 71ms/step - loss: 0.1652 - accuracy: 0.9458 - val_loss: 0.2529 - val_accuracy: 0.9180 - lr: 5.0000e-05
Epoch 163/200
45/46 [=====>.] - ETA: 0s - loss: 0.1738 - accuracy: 0.9417
Epoch 163: val_accuracy did not improve from 0.94262
46/46 [=====] - 3s 74ms/step - loss: 0.1757 - accuracy: 0.9403 - val_loss: 0.2428 - val_accuracy: 0.9344 - lr: 5.0000e-05
Epoch 164/200
45/46 [=====>.] - ETA: 0s - loss: 0.1771 - accuracy: 0.9431
Epoch 164: val_accuracy did not improve from 0.94262

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46/46 [=====] - 3s 72ms/step - loss: 0.1788 - accuracy:
0.9430 - val_loss: 0.2343 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 165/200
45/46 [=====>.] - ETA: 0s - loss: 0.1627 - accuracy: 0.9438
Epoch 165: val_accuracy improved from 0.94262 to 0.94536, saving model to /kaggle/
working/CASA_IRB_165_0.95.h5
46/46 [=====] - 3s 74ms/step - loss: 0.1630 - accuracy:
0.9437 - val_loss: 0.2386 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 166/200
45/46 [=====>.] - ETA: 0s - loss: 0.1559 - accuracy: 0.9514
Epoch 166: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1581 - accuracy:
0.9492 - val_loss: 0.2725 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 167/200
45/46 [=====>.] - ETA: 0s - loss: 0.1720 - accuracy: 0.9396
Epoch 167: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1710 - accuracy:
0.9403 - val_loss: 0.2283 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 168/200
45/46 [=====>.] - ETA: 0s - loss: 0.1602 - accuracy: 0.9493
Epoch 168: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1594 - accuracy:
0.9499 - val_loss: 0.2387 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 169/200
45/46 [=====>.] - ETA: 0s - loss: 0.1563 - accuracy: 0.9486
Epoch 169: val_accuracy did not improve from 0.94536
46/46 [=====] - 4s 77ms/step - loss: 0.1559 - accuracy:
0.9485 - val_loss: 0.2389 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 170/200
45/46 [=====>.] - ETA: 0s - loss: 0.1639 - accuracy: 0.9438
Epoch 170: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1641 - accuracy:
0.9430 - val_loss: 0.2320 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 171/200
45/46 [=====>.] - ETA: 0s - loss: 0.1721 - accuracy: 0.9424
Epoch 171: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1736 - accuracy:
0.9423 - val_loss: 0.2777 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 172/200
45/46 [=====>.] - ETA: 0s - loss: 0.1589 - accuracy: 0.9451
Epoch 172: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1579 - accuracy:
0.9458 - val_loss: 0.2796 - val_accuracy: 0.8962 - lr: 5.0000e-05
Epoch 173/200
45/46 [=====>.] - ETA: 0s - loss: 0.1750 - accuracy: 0.9458
Epoch 173: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 71ms/step - loss: 0.1752 - accuracy:
0.9458 - val_loss: 0.2398 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 174/200
45/46 [=====>.] - ETA: 0s - loss: 0.1650 - accuracy: 0.9438
Epoch 174: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1637 - accuracy:
0.9444 - val_loss: 0.2824 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 175/200
45/46 [=====>.] - ETA: 0s - loss: 0.1543 - accuracy: 0.9486
Epoch 175: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1550 - accuracy:
0.9478 - val_loss: 0.2455 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 176/200
45/46 [=====>.] - ETA: 0s - loss: 0.1844 - accuracy: 0.9382
Epoch 176: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1833 - accuracy:
0.9389 - val_loss: 0.2287 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 177/200
```

45/46 [=====>.] - ETA: 0s - loss: 0.1535 - accuracy: 0.9410
Epoch 177: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1534 - accuracy: 0.9410 - val_loss: 0.2160 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 178/200
45/46 [=====>.] - ETA: 0s - loss: 0.1710 - accuracy: 0.9424
Epoch 178: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 75ms/step - loss: 0.1695 - accuracy: 0.9430 - val_loss: 0.2232 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 179/200
45/46 [=====>.] - ETA: 0s - loss: 0.1577 - accuracy: 0.9444
Epoch 179: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 75ms/step - loss: 0.1634 - accuracy: 0.9430 - val_loss: 0.2355 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 180/200
45/46 [=====>.] - ETA: 0s - loss: 0.1615 - accuracy: 0.9458
Epoch 180: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1659 - accuracy: 0.9451 - val_loss: 0.3069 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 181/200
45/46 [=====>.] - ETA: 0s - loss: 0.1713 - accuracy: 0.9410
Epoch 181: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 75ms/step - loss: 0.1721 - accuracy: 0.9410 - val_loss: 0.2426 - val_accuracy: 0.9235 - lr: 5.0000e-05
Epoch 182/200
45/46 [=====>.] - ETA: 0s - loss: 0.1628 - accuracy: 0.9396
Epoch 182: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1621 - accuracy: 0.9396 - val_loss: 0.4175 - val_accuracy: 0.8361 - lr: 5.0000e-05
Epoch 183/200
45/46 [=====>.] - ETA: 0s - loss: 0.1555 - accuracy: 0.9493
Epoch 183: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1549 - accuracy: 0.9492 - val_loss: 0.3428 - val_accuracy: 0.8743 - lr: 5.0000e-05
Epoch 184/200
45/46 [=====>.] - ETA: 0s - loss: 0.1637 - accuracy: 0.9465
Epoch 184: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1647 - accuracy: 0.9458 - val_loss: 0.2864 - val_accuracy: 0.9071 - lr: 5.0000e-05
Epoch 185/200
45/46 [=====>.] - ETA: 0s - loss: 0.1694 - accuracy: 0.9479
Epoch 185: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1678 - accuracy: 0.9485 - val_loss: 0.2431 - val_accuracy: 0.9372 - lr: 5.0000e-05
Epoch 186/200
45/46 [=====>.] - ETA: 0s - loss: 0.1522 - accuracy: 0.9521
Epoch 186: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 74ms/step - loss: 0.1516 - accuracy: 0.9520 - val_loss: 0.2233 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 187/200
45/46 [=====>.] - ETA: 0s - loss: 0.1751 - accuracy: 0.9417
Epoch 187: val_accuracy did not improve from 0.94536
46/46 [=====] - 3s 72ms/step - loss: 0.1736 - accuracy: 0.9423 - val_loss: 0.2411 - val_accuracy: 0.9290 - lr: 5.0000e-05
Epoch 188/200
45/46 [=====>.] - ETA: 0s - loss: 0.1653 - accuracy: 0.9472
Epoch 188: val_accuracy improved from 0.94536 to 0.94809, saving model to /kaggle/working/CASA_IRB_188_0.95.h5
46/46 [=====] - 4s 77ms/step - loss: 0.1659 - accuracy: 0.9472 - val_loss: 0.2140 - val_accuracy: 0.9481 - lr: 5.0000e-05
Epoch 189/200
45/46 [=====>.] - ETA: 0s - loss: 0.1676 - accuracy: 0.9500
Epoch 189: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1664 - accuracy:

```

0.9506 - val_loss: 0.2919 - val_accuracy: 0.8989 - lr: 5.0000e-05
Epoch 190/200
45/46 [=====>.] - ETA: 0s - loss: 0.1646 - accuracy: 0.9451
Epoch 190: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1631 - accuracy:
0.9458 - val_loss: 0.2508 - val_accuracy: 0.9153 - lr: 5.0000e-05
Epoch 191/200
45/46 [=====>.] - ETA: 0s - loss: 0.1372 - accuracy: 0.9542
Epoch 191: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.1402 - accuracy:
0.9533 - val_loss: 0.2113 - val_accuracy: 0.9426 - lr: 5.0000e-05
Epoch 192/200
45/46 [=====>.] - ETA: 0s - loss: 0.1586 - accuracy: 0.9528
Epoch 192: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.1613 - accuracy:
0.9520 - val_loss: 0.2499 - val_accuracy: 0.9317 - lr: 5.0000e-05
Epoch 193/200
45/46 [=====>.] - ETA: 0s - loss: 0.1533 - accuracy: 0.9507
Epoch 193: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.1523 - accuracy:
0.9513 - val_loss: 0.2384 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 194/200
45/46 [=====>.] - ETA: 0s - loss: 0.1576 - accuracy: 0.9465
Epoch 194: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1578 - accuracy:
0.9465 - val_loss: 0.2750 - val_accuracy: 0.9208 - lr: 5.0000e-05
Epoch 195/200
45/46 [=====>.] - ETA: 0s - loss: 0.1446 - accuracy: 0.9535
Epoch 195: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1439 - accuracy:
0.9533 - val_loss: 0.2146 - val_accuracy: 0.9454 - lr: 5.0000e-05
Epoch 196/200
45/46 [=====>.] - ETA: 0s - loss: 0.1512 - accuracy: 0.9486
Epoch 196: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 75ms/step - loss: 0.1522 - accuracy:
0.9478 - val_loss: 0.2325 - val_accuracy: 0.9399 - lr: 5.0000e-05
Epoch 197/200
45/46 [=====>.] - ETA: 0s - loss: 0.1596 - accuracy: 0.9493
Epoch 197: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.1615 - accuracy:
0.9478 - val_loss: 0.2713 - val_accuracy: 0.9044 - lr: 5.0000e-05
Epoch 198/200
45/46 [=====>.] - ETA: 0s - loss: 0.1584 - accuracy: 0.9507
Epoch 198: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 76ms/step - loss: 0.1571 - accuracy:
0.9513 - val_loss: 0.2960 - val_accuracy: 0.8852 - lr: 5.0000e-05
Epoch 199/200
45/46 [=====>.] - ETA: 0s - loss: 0.1672 - accuracy: 0.9396
Epoch 199: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 74ms/step - loss: 0.1675 - accuracy:
0.9396 - val_loss: 0.3111 - val_accuracy: 0.8825 - lr: 5.0000e-05
Epoch 200/200
45/46 [=====>.] - ETA: 0s - loss: 0.1511 - accuracy: 0.9493
Epoch 200: val_accuracy did not improve from 0.94809
46/46 [=====] - 3s 72ms/step - loss: 0.1510 - accuracy:
0.9492 - val_loss: 0.2762 - val_accuracy: 0.9208 - lr: 5.0000e-05

```

In [160...

```

print(history.history.keys())

# "Accuracy"
plt.figure(figsize=(8, 5))
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Accuracy vs Epochs')

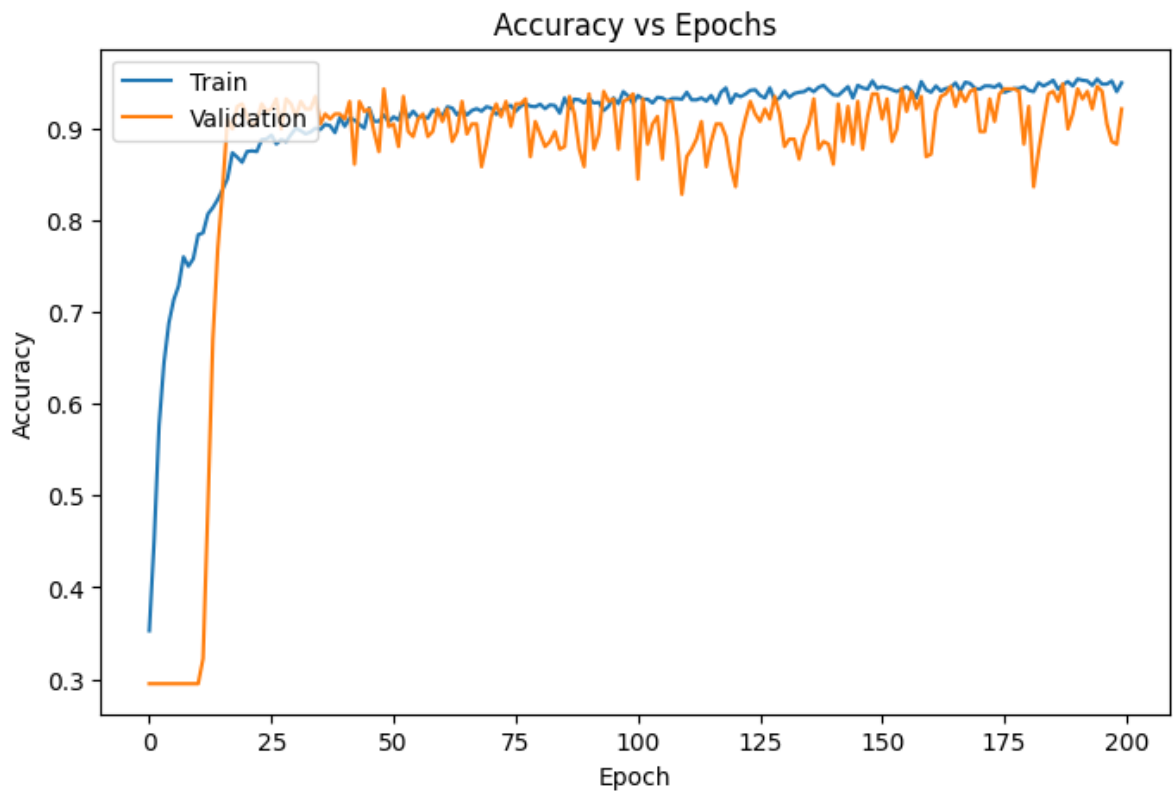
```

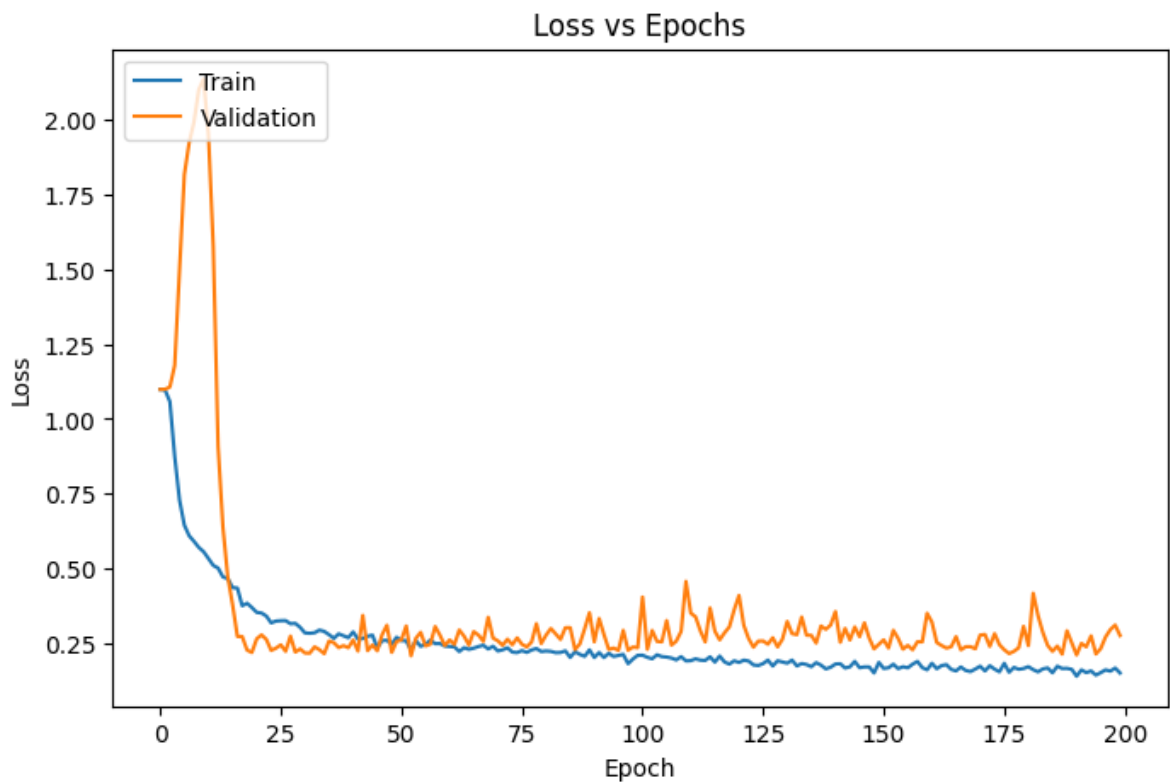


```
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left')
#plt.savefig('acc_97_depthwiseCaSa.png', dpi=1200)
plt.show()

# "Loss"
plt.figure(figsize=(8, 5))
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Loss vs Epochs')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left')
#plt.savefig('loss_97_depthwiseCaSa.png', dpi=1200)
plt.show()
```

```
dict_keys(['loss', 'accuracy', 'val_loss', 'val_accuracy', 'lr'])
```





Performance Evaluation

```
In [161...] re_model = tf.keras.models.load_model('/kaggle/working/CASA_IRB_188_0.95.h5',
                                              custom_objects={'ChannelAttentionModule':ChannelAt
                                                              'SpatialAttentionModule':SpatialAt
```

```
In [166...] re_model.evaluate(X_test, y_test)
```

```
12/12 [=====] - 1s 22ms/step - loss: 0.2140 - accuracy: 0.9481
```

```
Out[166]: [0.21397647261619568, 0.9480874538421631]
```

```
In [167...] y_pred = re_model.predict(X_test)
```

```
12/12 [=====] - 0s 21ms/step
```

```
In [168...] y_pred = np.argmax(y_pred, axis=1)
y_pred
```

```
Out[168]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
2, 2, 2, 2, 2, 0, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 0, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 1, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 1, 1, 2, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 0, 2, 2, 2, 2, 2, 2, 2, 2, 2,
2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 2, 2, 2])
```

In [169... `y_true = y_test`

In [170... `print(classification_report(y_true, y_pred))`

	precision	recall	f1-score	support
0	0.95	0.97	0.96	108
1	0.92	0.97	0.95	134
2	0.97	0.90	0.94	124
accuracy			0.95	366
macro avg	0.95	0.95	0.95	366
weighted avg	0.95	0.95	0.95	366

In []: